Ayush Vora

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Education

University of British Columbia, B.Sc, Computer Science and Statistics Major — Sep 2021 – May 2026

- **Cumulative Average:** 86.1%, 3.98/4.33 GPA
- **Relevant Courses:** Advanced Algorithm Design, Randomized Algorithms, Computational Theory, Machine Learning, Quantum Computation Statistical Inference, Time-series Analysis, Databases.

Interests

Algorithms, CS Education, Theoretical Computer Science, Randomized Algorithms, Communication, Computational Geometry, Software Engineering, Data Structures, Machine Learning, Data Science, Data Analysis, Statistical Inference.

Written Work

The Moment-Matching Technique and it's Application — Dec 2024

- **Authors:** Ayush Vora, supervised by Nathan Harms
- **Link:** [GitHub]
- Surveyed and explained the Moment Matching technique by summarizing Raskhodnikova et al. (2009), focusing on how it is used to prove lower bounds for approximating the Distribution Support Size problem.
- Demonstrated how Moment Matching naturally arises in randomized algorithm analysis and showed how it can be applied both to the Distribution Support Size problem and to related problems in probabilistic computation.
- Presented the material in a lecture-note style, distilling complex theoretical ideas into accessible explanations intended to teach the technique to students or researchers new to the area.

Dequantization of Quantum Recommendation Systems — Dec 2024

- **Authors:** Ayush Vora, Joshua Ho, Paige Ingram
- **Link:** [Google Drive]
- Explains the quantum recommendation system algorithm of Kerenidis & Prakash, breaking down its data structures, singular value estimation, and quantum projection steps into an accessible, course-friendly exposition.
- Shows how to dequantize the quantum algorithm by replacing its strong input assumptions (like efficient quantum state preparation and QRAM) with equivalent classical sampling-and-query access assumptions, following Tang's framework. This demonstrates that the supposed exponential quantum speedup collapses to only a polynomial slowdown.
- Presents a full classical recommendation system algorithm (based on the MODFKV method) that efficiently produces low-rank approximations and samples good recommendations—illustrating that classical algorithms can match the quantum algorithm's performance under fair assumptions.

Teaching Experience

Teaching Assistant, UBC - CPSC 121: Models of Computation — Jan 2026 – May 2026

Teaching Assistant, UBC - CPSC 421: Introduction to Theory of Computing — Sep 2025 – Jan 2026

Teaching Assistant, UBC - CPSC 221: Basic Algorithms and Data Structures — Jan 2025 – May 2025

Teaching Assistant, UBC - CPSC 110: Computation, Programs, and Programming — Sep 2024 – Jan 2025

Secondary School Tutor, Mayfield SS - Computer Science 12, Math 12, Calculus 12 — Jan 2022 – Jan 2023

Professional Experience

Systems Administrator Co-op, BC Centre for Ability Jan 2025 - Jan 2026

- Automated archival and optimization of company's Salesforce database, clearing out 520 GB of storage for future use.
- Mass-deployed MS Office 365 to 100+ computers, involving organizing appointments with clients, creating automation scripts, and presenting a demonstration of new features.
- Assisted 100+ employees with miscellaneous IT needs, including hardware replacements, software installation, and other troubleshooting issues.

Toolset Software Developer / Data Analyst Co-op, ATS Corporation Jan 2024 - Apr 2024

- Implemented MS Excel Macros using Visual Basics for Applications to automate tasks and generate visual graphics for data, saving 150 hours of repetitive work yearly.
- Expanded and improved on previous iterations of Excel Macros to increase clarity of current data, leading to more impactful analysis of key performance indicators of company success.
- Created and automated a data pipeline to inject data into a Databricks SQL server, allowing for improved efficiency of KPI dashboard generation. Through Databricks, set the capabilities to implement a machine learning model for data analysis.
- Presented data findings to the Project Management team to report strengths, weaknesses, and opportunities for improvement based on KPI analytics.
- Fixed 12 bugs present in the Java-based report-generator METRIX, improving app efficiency and increasing productivity by 33%.

IT Technician, Rootnet Solutions May 2022 – Sep 2022

- Deployed 200+ laptops and desktops successfully to 5 Air Canada offices, including YYZ International Airport.
- Served as the primary point of contact for Air Canada YYZ Branch for IMAC needs.

Projects

Wakey Wakey! (WIP) github.com: smart-alarm

- Engineered a smart alarm clock using a Raspberry Pi 4, involving modelling and 3D printing a shell and case for the display and stand.
- Programmed a back-end API and cloud-hosted NO-SQL database to update and store user data, saved alarms, settings, and more. The physical clock polls from the database to stay updated often.
- Designed a front-end application where the user can set alarms, create goals, and change user settings by communicating to the database.
- Tools used: Raspberry Pi 4, Linux, MongoDB, Python (FastAPI), JavaScript (React Native), Bambu Studio.

SpotiPaw github.com: spoti-paw

- Developed a motion-controlled media player for Spotify, which detects hand gestures and controls music accordingly
- Created with an ultrasound sensor connected to an Arduino, which connects to a Flask server managing the Spotify API.
- Tools Used: C (Arduino), Python (Flask), Spotify API.

SNASA github.com: SNASA

- Created an accessible website where the user can view different space-related objects, such as galaxies, planets, and satellites.
- Used SQL to create different queries, including adding, removing, and updating current objects, and different views of current systems.
- Tools Used: PHP, SQL (Oracle), HTML, JS

Skills

JavaScript (React/React Native, Node.js), **Python** (FastAPI, Flask), **Java** (Android Studio, Swing), **C** (Arduino), **C++**, **C#** (Unity), **R** (Tidyverse, Jupyter), **SQL** (Oracle, MySQL, Databricks), **No-SQL** (MongoDB, Firebase), HTML, CSS, Raspberry Pi, **Office 365** (VBA, Excel, Word, PowerBI), Problem Solving, Critical Thinking, Adaptability, Leadership, Teamwork, Time Management